

THE DIMENSION FORMULA FOR SELF-SIMILAR MEASURES WITH VIRTUALLY SOLVABLE ROTATIONS

GPT-6 ASTRA

ABSTRACT. We extend the argument of [KK26] to finitely supported, strictly contracting similarity systems with virtually solvable rotations. The dimension is expressed through entropy rates on invariant quotients. Full-dimensional conditional measures along proper invariant subspaces are the obstruction to the usual dimension formula.

This is an unedited model output on the basis of the paper [KK26] with the prompting done by C.K. and using the prompters local context.

1. STATEMENTS AND THE WEAKER HYPOTHESIS

The main theorem of [KK26] states that every self-similar measure ν arising from finitely many contracting similarities of the real line satisfies

$$\dim \nu = \min\{1, h_\mu/|\chi_\mu|\}.$$

Here h_μ is the entropy rate of random compositions, with coincident maps identified, and $\chi_\mu < 0$ is the mean logarithmic contraction. No separation or algebraicity assumption is required. We extend this formula to higher dimensions, where invariant subspaces can create an additional obstruction.

Let

$$g_i(x) = \rho_i U_i x + b_i \quad (1 \leq i \leq k), \quad 0 < \rho_i < 1, \quad U_i \in O(d), \quad b_i \in \mathbb{R}^d,$$

be distinct similarities, let $\mu = \sum_i p_i \delta_{g_i}$ with $p_i > 0$, and let $\nu = \sum_i p_i (g_i)_* \nu$. Denote the attractor by K . Except where explicitly stated otherwise, assume $\text{aff } K = \mathbb{R}^d$. This is affine irreducibility. For systems without a common fixed point, it is weaker than irreducibility of the rotations. Put $\Gamma = \langle U_1, \dots, U_k \rangle$.

For independent μ -distributed increments write

$$G_n = X_1 \cdots X_n = (A_n, B_n), \quad \lambda = - \sum_i p_i \log \rho_i > 0,$$

$$h = \lim_n \frac{H(G_n)}{n}, \quad h_{\text{lin}} = \lim_n \frac{H(A_n)}{n}, \quad \alpha = \dim \nu.$$

Here (A, B) denotes the map $x \mapsto Ax + B$; for a general similarity g write $A(g) = \rho(g)U(g)$ and $b(g)$ for these parts. Entropies use natural logarithms and identify exact coincidences of maps. Both entropy rates exist by subadditivity. Exact dimensionality is supplied by [FH09].

If V is Γ -invariant, its orthogonal complement is invariant too. Let $\pi_V : \mathbb{R}^d \rightarrow V^\perp$ be orthogonal projection, and write ν_x^V for the conditional measure on $x + V$.

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These conditionals are almost surely exact dimensional, with constant dimension, and

$$\dim \nu_x^V = \alpha - \dim(\pi_V \nu) \quad \text{for } \nu\text{-almost every } x \quad (1.1)$$

by [Hoc17, Theorem 6.13]. Call V *saturated* when this dimension is $\dim V$. The zero subspace is always saturated.

The maps induce similarities on $V^\perp$,

$$g_{i,V}(y) = \rho_i(U_i|_{V^\perp})y + \pi_V b_i.$$

Let h_V be the random-walk entropy rate of their law, merging coincident maps. Thus $h_{\{0\}} = h$, and set $h_{\mathbb{R}^d} = 0$.

Theorem 1.1 (Dimension through invariant quotients). *Suppose $H(A_n) = o(n)$. Then*

$$\dim \nu = \min_{V \text{ is } \Gamma\text{-invariant}} \left(\dim V + \frac{h_V}{\lambda} \right). \quad (1.2)$$

The minimum is attained at a saturated invariant subspace. In particular this holds whenever Γ is virtually solvable.

The substantive extension of the variance argument is the following obstruction theorem; it also records the cost of positive linear entropy.

Theorem 1.2 (Invariant-subspace obstruction). *Without any group-theoretic assumption on Γ , if*

$$\alpha < \min\{d, (h - h_{\text{lin}})/\lambda\}, \quad (1.3)$$

then there is a nonzero proper Γ -invariant subspace V such that

$$\dim \nu_x^V = \dim V \quad \text{for } \nu\text{-almost every } x. \quad (1.4)$$

Corollary 1.3 (Hochman's weaker condition). *Suppose $H(A_n) = o(n)$, in particular if Γ is virtually solvable. If no nonzero proper invariant subspace is saturated, then*

$$\dim \nu = \min\{d, h/\lambda\}.$$

Equivalently, the required condition is that for every such V ,

$$\dim \nu_x^V < \dim V \quad \text{almost surely,} \quad \text{or} \quad \alpha < \dim V + \dim(\pi_V \nu). \quad (1.5)$$

Ordinary linear irreducibility is a sufficient special case.

Thus the condition concerns dimensions of *conditional measures on fibers*, not dimensions of projections alone. It is precisely the exclusion of Hochman's invariant-subspace alternative. It is a sufficient condition, not a necessary one: the usual formula can hold even if a proper saturated subspace exists. Formula (1.2) also covers that case. The obstruction theorem concerns $\alpha < d$ and retains affine irreducibility; affine irreducibility alone does not preclude a dimension drop.

Corollary 1.4 (Self-similar sets). *Suppose Γ is virtually solvable and the maps have no exact overlaps. Let $\sum_i \rho_i^s = 1$. Then either $\dim_H K = \min\{d, s\}$, or there is a nonzero proper invariant V with*

$$\dim_H(K \cap (x + V)) = \dim V$$

for almost every x with respect to the self-similar measure of weights $p_i = \rho_i^s$.

The proof uses the scalar cumulative-entropy inequality [KK26, Lemma 2.1], Hochman's fine-scale entropy theorem, and his entropy formula for invariant fibers [Hoc17, Theorems 1.5 and 6.11]. The vector entropy inequality and the modified variance summation are proved below. The latter extends the approach of [KK25, Sections 5–6] by retaining an invariant covariance matrix instead of obtaining variance in every direction. No separation hypothesis is imposed. We retain strict contraction; contraction on average would require additional stopping and tail estimates.

2. LINEAR PARTS AND THE ENTROPY GAP

Lemma 2.1. *If $\Gamma \leq O(d)$ is finitely generated and virtually solvable, it is virtually abelian and has polynomial word growth. Consequently, for the system above, $H(A_n) = O(\log(n+1))$.*

Proof. Choose a solvable subgroup Γ_0 of finite index. Its closure L in $O(d)$ is solvable: continuity of commutator words preserves any fixed bound on the derived length. The identity component L^0 is a connected compact solvable Lie group, hence abelian. For completeness, Lie's theorem triangularizes its complexified infinitesimal representation; invariance of orthogonal complements for a unitary representation makes this flag an orthogonal decomposition into one-dimensional invariant spaces. Connectedness then makes L^0 simultaneously diagonalizable. As a compact Lie group, L has finitely many components. Thus $\Gamma_0 \cap L^0$ is abelian of finite index in Γ .

Taking its normal core gives a finite-index normal abelian subgroup F . Choose representatives of Γ/F . On multiplying a representative by one of the finitely many generators or their inverses, the correction in F belongs to a fixed finite set. A word of length at most n is therefore a product of at most n such corrections followed by one representative. The corrections commute, so there are at most $C(n+1)^q$ possible products, for fixed C, q . The scalar contraction of G_n is determined by the k letter counts, with at most $(n+1)^k$ possibilities. Hence A_n has polynomially many values and the entropy assertion follows. $\square$

Only this implication from virtual solvability is needed. Thus the group reduction does not require a quantitative Tits alternative. Also $H(\rho(G_n)) = O(\log(n+1))$ implies $h_{\text{lin}} = \lim_n H(U(G_n))/n$.

For a bounded random vector Y define averaged cube entropy by

$$H(Y; r) = \frac{1}{r^d} \int_{[0, r]^d} H\left(\left\lfloor \frac{Y+t}{r} \right\rfloor\right) dt, \quad H(Y; r \mid 2r) = H(Y; r) - H(Y; 2r).$$

The floor is coordinatewise. The first functional is concave and translation invariant. The second is also concave, translation invariant, nonnegative, and at most $d \log 2$: average conditional entropies for nested grids over $[0, 2r)^d$. These facts also imply that adding an independent summand cannot decrease entropy between these two meshes.

Lemma 2.2 (Hochman's entropy estimate). *Assume $\alpha < \min\{d, h/\lambda\}$ and that no nonzero proper invariant subspace is saturated. Then, for every fixed $C > e^\lambda$,*

$$H(B_n; C^{-n}) = n\lambda\alpha + o(n). \quad (2.1)$$

The same holds with any fixed constant multiple of the mesh.

Proof. At the natural dyadic level $\ell_n = \lfloor n\lambda/\log 2 \rfloor$,

$$H(B_n, \mathcal{D}_{\ell_n}) = n\lambda\alpha + o(n). \quad (2.2)$$

Indeed couple $Z \sim \nu$ as $Z = B_n + A_n Z'$ with an independent stationary tail Z' . The tail is bounded, and the law of large numbers gives $\log \|A_n\| = -n\lambda + o(n)$ in probability. On the event with error at most εn , the two natural-scale cube labels differ by at most $O(e^{\varepsilon n})$ in each coordinate, costing $O_d(\varepsilon n)$ entropy. On its complement, bounded support gives an $O(n)$ entropy cost multiplied by a probability tending to zero. Compare with the entropy dimension of ν , and then let $\varepsilon \downarrow 0$.

Apply [Hoc17, Theorem 1.5] to the original IFS. Its dimension-equality alternative fails because $h \leq H(\mu)$ and $\alpha < \min\{d, h/\lambda\}$. The hypothesis excludes the proper invariant-subspace alternative, and $\alpha < d$ excludes saturation of $\mathbb{R}^d$. The remaining alternative says that the entropy of the random similarity at any fixed exponentially finer group resolution, conditional on its translation at the natural scale, is $o(n)$. Choose a group resolution finer than both C^{-n} and the natural scale and project to translations. Together with (2.2), this gives the upper bound in (2.1); refinement gives the lower bound. Moving the origin of a cube grid changes entropy by at most $d \log 2$. Comparable meshes also have bounded comparison error, proving the assertions for averaged entropy and fixed mesh multiples. This use of Hochman's theorem allows exact overlaps. $\square$

3. A BOUNDED-COMPONENT VARIANCE ENERGY

We write $\text{Cov}(Y) = \mathbb{E}[(Y - \mathbb{E}Y)(Y - \mathbb{E}Y)^T]$; $M \preceq N$ means that $N - M$ is positive semidefinite.

For a compactly supported probability measure λ on $\mathbb{R}^d$, define

$$\mathcal{V}_\lambda(r) = r^{-2} \sup_{\lambda = \int \lambda_z d\theta(z)} \int \text{tr Cov}(\lambda_z) d\theta(z), \quad \text{diam supp } \lambda_z \leq r,$$

and $\mathcal{E}_R(\lambda) = \int_0^R \mathcal{V}_\lambda(r) dr/r$, $\mathcal{E} = \mathcal{E}_\infty$. Point-mass disintegrations are allowed. For finite laws, the supremum is attained with finitely many labels: merge components having the same support subset, which can only increase average covariance trace, and optimize their atom-mass allocations on a compact polytope. Thus all later maximizing choices can be made measurably, since each conditioned block has only finitely many possible laws.

Lemma 3.1. *The functional $\mathcal{V}_\lambda(r)$ is concave in λ , and $0 \leq \mathcal{V}_\lambda(r) \leq 1/2$. If g is a similarity of ratio $a > 0$, then $\mathcal{V}_{g\lambda}(r) = \mathcal{V}_\lambda(r/a)$ and $\mathcal{E}(g\lambda) = \mathcal{E}(\lambda)$. If $\text{diam supp } \lambda \leq D$, then, for $R \geq D$,*

$$\mathcal{E}(\lambda) - \mathcal{E}_R(\lambda) = \frac{\text{tr Cov}(\lambda)}{2R^2} \leq \frac{D^2}{4R^2}.$$

For a finite law, $\mathcal{E}$ is finite.

Proof. Mixing admissible disintegrations proves concavity. For an independent pair Z, Z' of law λ_z , $\text{tr Cov}(\lambda_z) = \frac{1}{2} \mathbb{E}|Z - Z'|^2 \leq r^2/2$. Similarity covariance is immediate. When $r \geq D$, the trivial one-component disintegration attains the full variance; the law of total variance gives the converse bound. Integrating $r^{-2} \text{tr Cov}(\lambda)$ proves the tail identity. Below the smallest positive inter-atom distance only point-mass components are allowed. $\square$

4. POISSON HYPERPLANES AND ADDITIVE ENTROPY LOSS

Let σ be normalized rotation-invariant measure on S^{d-1} , and put

$$c_d = \int |\langle \theta, e_1 \rangle| d\sigma(\theta) > 0.$$

Take a Poisson process on $S^{d-1} \times \mathbb{R} \times (0, \infty)$ with intensity $c_d^{-1} d\sigma(\theta) da dt$. At time t , the hyperplanes $\langle \theta, x \rangle = a$ partition $\mathbb{R}^d$; denote the partition by $\mathcal{P}_t$. Hyperplanes are parametrized with their two orientations; the stated intensity is chosen so that

$$\mathbb{P}\{x, y \text{ belong to the same cell of } \mathcal{P}_t\} = e^{-t|x-y|}. \quad (4.1)$$

The partition is always included in the conditioning. For finitely supported X put

$$v_X(t) = \mathbb{E} \operatorname{tr} \operatorname{Cov}(X \mid \mathcal{P}_t(X)), \quad \mathcal{E}_P(X) = \int_0^\infty t v_X(t) dt, \quad D(X) = \mathbb{E}|X - X'|.$$

Lemma 4.1. *For finitely supported X ,*

$$D(X) \leq 2 \int_0^\infty v_X(t) dt, \quad \int_0^\infty \mathbb{E} D(X \mid \mathcal{P}_t(X)) dt \leq 2 \mathcal{E}_P(X).$$

Proof. For each fixed partition, the independent-pair identity gives

$$\mathbb{E}[|X - X'|^2 \mathbf{1}_{\{\mathcal{P}_t(X) = \mathcal{P}_t(X')\}}] \leq 2v_X(t).$$

Average over the partition and use (4.1). The integral of $u^2 e^{-tu}$ in t is u , yielding the first assertion. Apply the first assertion to each conditional cell law at time t . Further independent hyperplanes of intensity s yield the partition at time $t + s$, hence

$$\mathbb{E} D(X \mid \mathcal{P}_t(X)) \leq 2 \int_t^\infty v_X(s) ds.$$

Tonelli proves the second assertion. $\square$

Lemma 4.2. *If $X_1, \dots, X_m$ are independent finitely supported random vectors, then*

$$\sum_i H(X_i) - H\left(\sum_i X_i\right) \leq 18m \sum_i \mathcal{E}_P(X_i).$$

Proof. Use independent hyperplane processes for the inputs, put $S = \sum_i X_i$, and let

$$J(t) = \mathbb{E} H(\mathcal{P}_t^1(X_1), \dots, \mathcal{P}_t^m(X_m) \mid S).$$

As $t \rightarrow \infty$, all atoms are distinguished almost surely, so $J(\infty) = \sum_i H(X_i) - H(S)$ and $J(0) = 0$. For a fixed tuple of current cells, the input variables remain independent. A cut of input i in direction θ reveals the bit $\mathbf{1}_{\{\langle \theta, X_i \rangle \leq a\}}$. Conditioning on S reveals at least its scalar projection. The scalar conditional cumulative-entropy lemma of [KK26, Lemma 2.1] therefore bounds the integral in a of this bit's conditional entropy by

$$9 \left(D(\langle \theta, X_i \rangle) + D\left(\sum_{j \neq i} \langle \theta, X_j \rangle\right) \right) \leq 9 \sum_j D(\langle \theta, X_j \rangle),$$

with the laws in this display conditioned on the current cell tuple. Integrating in $c_d^{-1}\sigma$ turns projected absolute dispersion into Euclidean dispersion. Summing the m possible input indices gives

$$J'(t) \leq 9m \sum_i \mathbb{E}D(X_i \mid \mathcal{P}_t^i(X_i)).$$

To justify differentiation, restrict to hyperplanes meeting the convex hull of a finite support; their total intensity is finite. The entropy chain rule and the one-jump formula for a Poisson process apply. Integrate and use Lemma 4.1. $\square$

Lemma 4.3. *There is a function $p_d(A) \rightarrow 0$ exponentially as $A \rightarrow \infty$ such that the cell of a fixed point in $\mathcal{P}_{t/2}$ has diameter exceeding A/t with probability at most $p_d(A)$, uniformly in the point and $t > 0$.*

Proof. By stationarity and scaling take the point to be zero and $t = 1$. Choose small spherical caps about each of $\pm e_j$, small enough that a choice of one unit normal from each cap positively spans $\mathbb{R}^d$ with a uniform bound. If, for each cap, there is a hyperplane with its outward normal in that cap and its positive distance from zero at most cA , then the halfspaces containing zero bound a polytope of diameter at most A , provided $c = c(d) > 0$ is sufficiently small. The intensity of each such family of hyperplanes is a positive dimension-dependent constant times A . A union bound over the $2d$ caps proves the assertion. $\square$

Lemma 4.4. *For some finite C_d and every finitely supported X , $\mathcal{E}_P(X) \leq C_d \mathcal{E}(X)$.*

Proof. Write $\mathcal{P}_t$ as the common refinement of independent partitions $\mathcal{P}_{t/2}^{(1)}$ and $\mathcal{P}_{t/2}^{(2)}$. Call a final cell bad if its diameter exceeds A/t , and let $v_{\text{bad}}(t)$ be its contribution to $v_X(t)$. For a first-partition cell C center at the conditional mean m_C . If $C \cap D$ is bad, the second-partition cell D is bad as well. Thus the contribution of such intersections is bounded by

$$\int_C |x - m_C|^2 \mathbf{1}_{\{\text{diam } \mathcal{P}_{t/2}^{(2)}(x) > A/t\}} d\mathcal{L}(X)(x).$$

Average over the second partition and use Lemma 4.3, then over the first partition. This gives

$$v_{\text{bad}}(t) \leq p_d(A) v_X(t/2), \quad \int_0^\infty t v_{\text{bad}}(t) dt \leq 4p_d(A) \mathcal{E}_P(X).$$

Keep each good cell as a component and split every bad cell into point masses. This is an admissible diameter- A/t disintegration, so

$$\mathcal{V}_X(A/t) \geq \frac{t^2}{A^2} (v_X(t) - v_{\text{bad}}(t)).$$

Integration and the substitution $r = A/t$ yield

$$\mathcal{E}(X) \geq A^{-2} (1 - 4p_d(A)) \mathcal{E}_P(X).$$

Choose A so that $4p_d(A) < 1$. To justify finiteness before rearranging, let $p_* > 0$ be the least atom mass. The conditional-pair formula gives

$$v_X(t) \leq \frac{1}{2p_*} \sum_{x \neq y} p_x p_y |x - y|^2 e^{-t|x-y|}.$$

Thus $\mathcal{E}_P(X) < \infty$. $\square$

Theorem 4.5 (Vector entropy loss). *For every d there is $K_d < \infty$ such that, for independent finitely supported random vectors $X_1, \dots, X_m \in \mathbb{R}^d$,*

$$\sum_{i=1}^m H(X_i) - H\left(\sum_{i=1}^m X_i\right) \leq K_d m \sum_{i=1}^m \mathcal{E}(X_i).$$

Proof. Combine Lemmas 4.2 and 4.4. $\square$

5. THE BLOCK ENTROPY IMPLICATION

We next convert unresolved discrete entropy into variance below a prescribed scale.

Proposition 5.1. *Suppose $Y = \sum_{j=1}^L A_j D_j$, with deterministic linear maps A_j and independent digits D_j taking at most K values. For every integer $m \geq 1$ and $R > 0$,*

$$\mathcal{E}_R(Y) \geq \frac{H(Y) - H(Y; R/\sqrt{d}) - L(K-1)\log(m+1)/m}{K_d m} - \frac{1}{4}.$$

Proof. For m independent copies, their sum is determined by the multiplicities of digit values at each of the L positions, and hence has entropy at most $L(K-1)\log(m+1)$. Condition each copy on its own cube label in a common grid of mesh $R/\sqrt{d}$. Apply Theorem 4.5 to the conditional independent laws. Each component has diameter at most R ; its energy above R is at most $1/4$ by Lemma 3.1. Concavity bounds the average energy below R by $\mathcal{E}_R(Y)$. Average the grid origin and rearrange. $\square$

Proposition 5.2 (Conditional translation energy). *Suppose the entropy estimate (2.1) holds for every fixed $C > e^\lambda$, and*

$$\alpha = \dim \nu < d, \quad \delta = h - h_{\text{lin}} - \lambda\alpha > 0.$$

There is $\eta > 0$ such that, for every sufficiently large fixed C and all sufficiently large n ,

$$\mathbb{E}\mathcal{E}_{C^{-n}}(\mathcal{L}(B_n | A_n)) \geq \eta n.$$

Proof. By the assumed entropy estimate, for every fixed sufficiently large C ,

$$H(B_n; C^{-n}/\sqrt{d}) = n\lambda\alpha + o(n).$$

Choose N with $H(A_N)/N < h_{\text{lin}} + \delta/4$, and split the first n increments into blocks of length N and one shorter block. Let T record the exact linear part of every block. If the successive block maps are $x \mapsto C_j x + D_j$, then

$$B_n = D_1 + C_1 D_2 + \dots + (C_1 \dots C_{L-1}) D_L, \quad L = \lceil n/N \rceil.$$

Conditional on $T = (C_1, \dots, C_L)$, the D_j are independent, their matrix coefficients are fixed, and each D_j takes at most $K = k^N$ values. Furthermore,

$$H(B_n | T) = H(G_n | T) \geq nh - H(T) \geq n(h - h_{\text{lin}} - \delta/4) - O(N).$$

By concavity, their average conditional entropy at mesh $C^{-n}/\sqrt{d}$ is at most the unconditional entropy there. Thus the average gap in Proposition 5.1 is at least $\delta n/2$ for large n . Choose m fixed, depending on N, K, δ , so that $(K-1)\log(m+1)/m < \delta N/8$. Proposition 5.1 gives a positive linear energy bound conditional on T . Since A_n is determined by T , concavity allows removal of the extra conditioning. $\square$

6. STATIONARY AVERAGING OF COVARIANCE

Write $\rho_* = \min_i \rho_i$ and $\rho^* = \max_i \rho_i < 1$. Let $\mathcal{K}$ be the compact closure of Γ and define its covariance averaging operator

$$\mathcal{T}(M) = \int_{\mathcal{K}} U M U^T dm_{\mathcal{K}}(U) \quad (M \in \text{Sym}_d).$$

It preserves positivity and trace. Its image consists of the symmetric matrices fixed by every conjugation $M \mapsto U_i M U_i^T$. Consequently the range of each matrix in its image is Γ -invariant. The next lemma realizes this averaging using a stationary random word, including the squared contraction factors.

Lemma 6.1 (Stationary covariance averaging). *There is a randomized bounded stopping word P , of length at most L_0 , and a constant $c_P > 0$ such that*

$$\mathbb{E}[P_* \nu] = \nu, \quad \mathbb{E}[A(P) M A(P)^T] = c_P \mathcal{T}(M) \quad (M \in \text{Sym}_d). \quad (6.1)$$

The randomization is independent of the increments, which have law μ .

Proof. Equip Sym_d with its Frobenius inner product. Let E be the common fixed space of conjugation and $W = E^\perp$. The conjugation representation $\mathcal{R}$ on W has Haar average zero. If $W = \{0\}$, take the empty word, so $c_P = 1$.

Otherwise put $S = \text{span}_{\mathbb{R}}\{\mathcal{R}(U) : U \in \mathcal{K}\}$. Zero lies in the relative interior of the convex hull of these operators: a supporting linear functional nonnegative on all of them has Haar average zero, hence vanishes on all of them and thus on S . The semigroup of positive words in the U_i is dense in $\mathcal{K}$; indeed a closed subsemigroup of a compact group is a group. We may therefore choose finitely many words w_a such that zero is interior, relative to S , to the convex hull of $\mathcal{R}(U(w_a))$.

Let q_a be the probability of the word w_a . Multiplying each of these finitely many operators by the positive number $q_a \rho(w_a)^2$ preserves the property that their positive cone is S , and hence preserves the interior property at zero. For a common integer $N \geq \max_a |w_a|$, stop at $|w_a|$ if the initial word is w_a , and otherwise stop at N . Denote this bounded stopping rule by τ_a . On W ,

$$\mathbb{E}[\rho(G_{\tau_a})^2 \mathcal{R}(U(G_{\tau_a}))] = q_a \rho(w_a)^2 \mathcal{R}(U(w_a)) + E_{a,N}, \quad \|E_{a,N}\| \leq (\rho^*)^{2N}.$$

Every error belongs to S . For sufficiently large N , zero is still interior to the convex hull of these expectations. Choose convex weights making their average zero, draw the index a independently, and use τ_a . This gives a bounded stopped word P . On E , every conjugation is the identity; on W , its contraction-weighted average is zero. Thus (6.1) holds with $c_P = \mathbb{E}\rho(P)^2 > 0$ and $L_0 = N$. The stationarity assertion follows by iterating $\nu = \mu * \nu$ over the finite tree of stopped words, and then averaging the independent index. $\square$

For each N and scale s , choose a maximizing diameter- s disintegration of every finite law $\mathcal{L}(B_N | A_N)$. Denote the combined linear-part and component label by Q . Set

$$C_N(s) = s^{-2} \mathbb{E} \text{Cov}(B_N | Q), \quad a_N(s) = \text{tr } C_N(s) = \mathbb{E} \mathcal{V}_{\mathcal{L}(B_N | A_N)}(s).$$

In particular $0 \leq a_N(s) \leq 1/2$.

Proposition 6.2 (Embedding separated blocks). *Let $(N_j, s_j)_{j=1}^m$ satisfy $0 < s_1 < \dots < s_m$ and*

$$s_{j+1} \geq \rho_*^{-(N_j + L_0 + 1)} s_j \quad (j < m). \quad (6.2)$$

Put $a_j = a_{N_j}(s_j)$, $a = \sum_j a_j$, and

$$D = \sum_j \mathcal{T}(C_{N_j}(s_j)), \quad \text{tr } D = a.$$

For every $0 < u \leq s_1$, ν has a disintegration into laws

$$z_\omega + \sum_{j=1}^m Y_{j,\omega},$$

where the summands are independent and centered, with $|Y_{j,\omega}| \leq u$. If Σ_ω is their total covariance, there is a random symmetric matrix P_ω such that, for constants $c_-, c_+ > 0$ depending only on μ ,

$$c_- D \preceq P_\omega \preceq c_+ D, \quad \mathbb{E} \|u^{-2} \Sigma_\omega - P_\omega\|_F^2 \leq a/2. \quad (6.3)$$

Proof. Run one walk. Before block j , wait until its prefix contraction first falls to at most u/s_j , insert an independent word with the law of P from Lemma 6.1, and then take N_j ordinary increments. The contraction t_j immediately before the padding lies in $(\rho_* u/s_j, u/s_j]$. After this padding and block it is greater than $\rho_*^{N_j+L_0+1} u/s_j \geq u/s_{j+1}$. Thus the selected blocks do not overlap. All waiting times have deterministic bounds since $\rho^* < 1$.

Record the intervening gap words, padding words, exact linear parts of the selected blocks, and their chosen component labels. Also record an independent tail point of law ν . Given this environment, the selected translations D_j remain independent with their component laws. Indeed, admissibility of the gaps depends only on the recorded linear parts, so the joint weight of compatible selected words is a common gap factor times the product of their word probabilities and component-label probabilities. The stopped endpoint is exactly

$$z_\omega + \sum_j R_j(D_j - \mathbb{E}[D_j | Q_j]),$$

where R_j is the recorded linear prefix including the padding. Stationarity at the bounded randomized stopping time gives the required disintegration. The centered summands have norm at most u .

Put $M_j = u^{-2} R_j \text{Cov}(D_j | Q_j) R_j^T$, so $u^{-2} \Sigma = \sum_j M_j$ and $0 \preceq M_j \preceq I/2$. Use the forward filtration $\mathcal{F}_j$ revealing actual words and auxiliary labels through block j . Conditional on the past and the current gap, Lemma 6.1 gives

$$\mathbb{E}[M_j | \text{past and current gap}] = c_P \frac{t_j^2 s_j^2}{u^2} \mathcal{T}(C_{N_j}(s_j)).$$

The prefix rotation disappears because the averaged matrix is invariant. After averaging the gap, write $P_j = \mathbb{E}[M_j | \mathcal{F}_{j-1}]$. Then

$$c_P \rho_*^2 \mathcal{T}(C_{N_j}(s_j)) \preceq P_j \preceq c_P \mathcal{T}(C_{N_j}(s_j)).$$

Thus $P_\omega = \sum_j P_j$ satisfies the first assertion in (6.3). Martingale orthogonality in the Frobenius space and $\text{tr}(M_j^2) \leq \frac{1}{2} \text{tr } M_j$ give

$$\mathbb{E} \left\| \sum_j (M_j - P_j) \right\|_F^2 \leq \sum_j \mathbb{E} \text{tr}(M_j^2) \leq \frac{1}{2} \sum_j a_j.$$

Here $\mathbb{E} \operatorname{tr} M_j \leq a_j$ since $\|R_j\| \leq u/s_j$. These are forward-filtration estimates, not estimates conditional on future labels. Their unconditional consequence is exactly (6.3) for the covariance of the disintegrated laws. $\square$

Lemma 6.3 (Arbitrarily large separated variance budgets). *Under the hypotheses of Proposition 5.2, for every $t > 0$ there is a finite list satisfying (6.2) with*

$$t \leq \sum_j a_j \leq t + \frac{1}{2}.$$

Proof. Fix $C > \rho_*^{-1}$ sufficiently large for Proposition 5.2. For large N , let

$$v_N(s) = \mathbb{E} \mathcal{V}_{\mathcal{L}(B_N|A_N)}(s), \quad D_N = (N + L_0 + 1) \log \rho_*^{-1}.$$

Averaging logarithmic grids $\{e^{v+jD_N} : j \in \mathbb{Z}\}$ over $0 \leq v < D_N$ and using the energy bound gives a finite packet of scales below C^{-N} , separated by e^{D_N} , whose total v_N is at least

$$\frac{\eta N}{D_N} \geq \eta_0 > 0.$$

Finiteness follows because these finite conditional laws have a positive minimum inter-atom distance. Having chosen a packet, choose the next N so large that

$$C^{-N} \rho_*^{-(N+L_0+1)} < \text{the smallest scale in all previously chosen packets.}$$

This is possible since $C\rho_* > 1$. Place the new packet before all earlier packets. Both the internal separations and the boundary separation then satisfy (6.2). After finitely many packets the total exceeds t . Read the final ordered list from its smallest scale and truncate immediately when the total first reaches t . Each summand is at most $1/2$, giving the claimed upper bound. $\square$

7. LARGE VARIANCE PRODUCES A SATURATED INVARIANT SUBSPACE

The next step replaces the full-dimensional central limit argument. It keeps directions in which variance accumulates and allows the limiting Gaussian to be singular on the ambient space.

Fix a subspace V of dimension q , and choose orthonormal coordinates adapted to $V \oplus V^\perp$. For $r > 0$ and an integer $m \geq 1$, take a nested pair of cube grids of sides r and $r2^{-m}$, with a common origin averaged uniformly over $[0, r]^d$. Let Q_f, Q_c be their labels and let $Q_f^\perp$ be the fine label in $V^\perp$. Define

$$F_{V,m}(\theta; r) = \mathbb{E}_{\text{origin}} H(Q_f(Z) \mid Q_c(Z), Q_f^\perp(Z)), \quad Z \sim \theta. \quad (7.1)$$

This is translation invariant and concave in θ , and $0 \leq F_{V,m} \leq qm \log 2$. Equivalently it is the difference between averaged fine-grid entropy and averaged entropy of the rectangular grid with side r in V and side $r2^{-m}$ in $V^\perp$.

Lemma 7.1 (Entropy continuity for small independent summands). *Fix V, m , a compact set $\mathcal{C}$ of positive semidefinite covariance matrices, and $\varepsilon > 0$. There is $\delta > 0$ such that the following holds. If the Y_j are independent and centered, $|Y_j| \leq \delta r$, and*

$$\operatorname{dist} \left(r^{-2} \sum_j \operatorname{Cov}(Y_j), \mathcal{C} \right) < \delta,$$

then, for some $Q \in \mathcal{C}$,

$$\left| F_{V,m} \left(\mathcal{L} \left(\sum_j Y_j \right); r \right) - F_{V,m}(\mathcal{N}(0, Q); 1) \right| < \varepsilon.$$

The matrices in $\mathcal{C}$ may be singular.

Proof. Scale to $r = 1$. If uniformity failed, there would be a sequence with maximal summand norm tending to zero and total covariance converging to some $Q \in \mathcal{C}$. Taylor expansion of characteristic functions proves convergence in law to $\mathcal{N}(0, Q)$: the total cubic error is at most $O(|\xi|^3 \delta B)$ and the quadratic logarithmic error at most $O(|\xi|^4 \delta^2 B)$, where B bounds total variance.

The two grid-label vectors have uniformly bounded second moments (with a bound depending on m). Their entropies are uniformly integrable in the labels: if p_k is an integer-vector distribution, $b = \sum_{|k| > R} p_k$, comparison with a law proportional to $(1 + |k|)^{-d-1}$ gives

$$\sum_{|k| > R} -p_k \log p_k \leq -b \log b + C_d b + (d+1) \mathbb{E}[\log(1 + |K|) \mathbf{1}_{\{|K| > R\}}].$$

The second-moment bound makes this uniformly small as $R \rightarrow \infty$. For almost every grid origin, every grid boundary has zero mass under the limiting Gaussian, including a singular Gaussian; this follows by integrating each boundary indicator over its shift. Thus weak convergence and the tail bound give convergence of both averaged grid entropies in (7.1), a contradiction. $\square$

Lemma 7.2 (Gaussian fiber entropy). *Suppose Q is supported on V and $Q|_V \succeq bI_V$ for $b > 0$. Then for every $m \geq 1$,*

$$F_{V,m}(\mathcal{N}(0, Q); 1) \geq qm \log 2 - \frac{q}{24b}.$$

Proof. For $G \sim \mathcal{N}(0, Q)$ the perpendicular label is deterministic, so the left side is $H(G|_V; 2^{-m}) - H(G|_V; 1)$. For an independent uniform cube U_s of side s in V , $H(G|_V; s) = h(G|_V + U_s) - q \log s$, where h is differential entropy. Entropy increase under independent addition and the Gaussian maximum-entropy inequality give

$$h(G|_V + U_{2^{-m}}) \geq h(G|_V),$$

$$h(G|_V + U_1) - h(G|_V) \leq \frac{1}{2} \log \det(I_V + (Q|_V)^{-1}/12) \leq \frac{q}{24b}.$$

Subtract. $\square$

Proposition 7.3 (Subspace variance criterion). *Suppose there are finite lists satisfying (6.2) with $\sum_j a_{N_j}(s_j)$ arbitrarily large. Then ν has a nonzero Γ -invariant saturated subspace, possibly all of $\mathbb{R}^d$.*

Proof. Choose lists indexed by ℓ with totals $a_\ell \rightarrow \infty$, and let D_ℓ be their averaged covariance matrices from Proposition 6.2. Passing to a subsequence,

$$D_\ell / a_\ell \rightarrow M, \quad M \succeq 0, \quad \text{tr } M = 1.$$

Each matrix is invariant under the rotations. Consequently $V = \text{Im } M$ is a nonzero invariant subspace. The compact matrix interval

$$\mathcal{C} = \{Q : c_- M \preceq Q \preceq c_+ M\}$$

consists of matrices supported on V , all uniformly positive definite on V . Set $b = c_- \lambda_{\min}(M|_V) > 0$ and $C_0 = \dim V / (24b)$.

Apply Proposition 6.2 to list ℓ at $u = r/\sqrt{a_\ell}$. This is allowed whenever r is sufficiently small for that fixed list. Its summands, divided by r , have norms at most $a_\ell^{-1/2}$. Moreover (6.3) gives

$$\mathbb{E} \left\| r^{-2} \Sigma - \frac{P_\omega}{a_\ell} \right\|_F^2 \leq \frac{1}{2a_\ell}, \quad c_- \frac{D_\ell}{a_\ell} \preceq \frac{P_\omega}{a_\ell} \preceq c_+ \frac{D_\ell}{a_\ell}.$$

The distance from this last matrix interval to $\mathcal{C}$ tends uniformly to zero: otherwise a bounded sequence in the intervals would have a limit outside $\mathcal{C}$, contradicting the limiting inequalities. Thus $\text{dist}(r^{-2}\Sigma, \mathcal{C}) \rightarrow 0$ in probability, uniformly in all sufficiently small r allowed by the list.

Fix m . Lemmas 7.1 and 7.2, followed by concavity of $F_{V,m}$, show that

$$\liminf_{r \downarrow 0} F_{V,m}(\nu; r) \geq (\dim V)m \log 2 - C_0. \quad (7.2)$$

Indeed choose ℓ large first, so the jump bound and the exceptional probability are as small as needed, and then allow every sufficiently small r . On exceptional components use $F_{V,m} \geq 0$. This order fixes V independently of both m and r .

It remains to pass from finite-resolution entropy to actual fibers; continuity of exact conditioning is not being assumed. Let $\mathcal{D}_n$ be the fixed dyadic grids in the adapted coordinates. Changing the origin of either grid entropy defining $F_{V,m}$ costs at most $d \log 2$. Consequently (7.2) implies

$$\liminf_{n \rightarrow \infty} H_\nu(\mathcal{D}_{n+m} \mid \mathcal{D}_n \vee \pi_V^{-1} \mathcal{D}_{n+m}^{V^\perp}) \geq (\dim V)m \log 2 - C_0 - 2d \log 2.$$

The conditional entropy here equals the average over level- n components of their entropy in the fine grid conditioned on the fine perpendicular label. By [Hoc17, Theorem 6.11], the lower and upper Cesaro limits in n , divided by $m \log 2$, both tend as $m \rightarrow \infty$ to the almost-sure value of $\dim \nu_x^V$. The displayed inequality gives $\dim \nu_x^V \geq \dim V$; the reverse inequality is automatic. Thus V is saturated. $\square$

8. CONCLUSION AND INVARIANT QUOTIENTS

For later use, the standard upper bound is

$$\alpha \leq \min\{d, h/\lambda\}. \quad (8.1)$$

Indeed apply the symbolic-entropy bound to the distinct maps in $\text{supp}(\mu^{*n})$, with their combined probabilities, to get $\alpha \leq H(G_n)/(n\lambda)$, and then let $n \rightarrow \infty$.

Proof of Theorem 1.2. Assume (1.3) and suppose no nonzero proper invariant subspace is saturated. Since $h_{\text{lin}} \geq 0$, Lemma 2.2 applies. Proposition 5.2 yields a linear conditional variance-energy bound. Lemma 6.3 supplies arbitrarily large separated lists, and Proposition 7.3 supplies a nonzero invariant saturated subspace V . It cannot be $\mathbb{R}^d$, since $\alpha < d$, and a proper such V contradicts the supposition. $\square$

Proof of Corollary 1.3. Combine Theorem 1.2, $h_{\text{lin}} = 0$, and (8.1). The equivalent form of the condition follows from (1.1). $\square$

Proof of Theorem 1.1. For every invariant V , dimension conservation and the entropy upper bound for the projected system give

$$\alpha \leq \dim V + \dim(\pi_V \nu) \leq \dim V + h_V/\lambda. \quad (8.2)$$

The case $V = \mathbb{R}^d$ means simply $\alpha \leq d$.

Choose a saturated invariant subspace V of maximum possible dimension; one exists because $\{0\}$ is saturated and dimensions take finitely many values. If $V = \mathbb{R}^d$, then $\alpha = d$ and equality holds in (8.2) for this V . Otherwise put $\bar{\nu} = \pi_V \nu$. This projected IFS is affinely irreducible, since $\text{aff } K = \mathbb{R}^d$, and its linear entropy is still $o(n)$. It has no nonzero saturated invariant subspace. In fact if $W \subseteq V^\perp$ were one, the invariant subspace $U = V \oplus W$ would satisfy, by two uses of dimension conservation,

$\dim \nu_x^U = \alpha - \dim(\pi_U \nu) = \dim V + \dim \bar{\nu} - \dim(\pi_W \bar{\nu}) = \dim V + \dim W = \dim U$, contradicting maximality of $\dim V$. Corollary 1.3 in the quotient therefore gives

$$\dim \bar{\nu} = \min\{d - \dim V, h_V/\lambda\}.$$

The first value cannot be attained, as that would make $V^\perp$ itself a nonzero saturated subspace of $\bar{\nu}$. Hence $\alpha = \dim V + \dim \bar{\nu} = \dim V + h_V/\lambda$. This proves both the minimum formula and its attainment. Finally Lemma 2.1 proves the virtually solvable assertion. $\square$

Proof of Corollary 1.4. Use $p_i = \rho_i^s$. Absence of exact overlaps gives $h = H(p) = s\lambda$. If $\dim_H K < \min\{d, s\}$, then $\dim \nu$ also falls below this bound, so Theorem 1.2 supplies a proper nonzero saturated invariant V . Almost every conditional is supported on $K \cap (x + V)$ and has dimension $\dim V$, proving the assertion. The matching upper bound $\dim_H K \leq \min\{d, s\}$ follows from cylinder coverings. $\square$

Remark 8.1 (Recovery of the irreducible theorem). For irreducible rotations the only invariant subspaces are 0 and $\mathbb{R}^d$, so (1.2) becomes $\min\{d, h/\lambda\}$. If affine irreducibility fails in this case, the attractor is a common fixed point. Then $B_n = x_0 - A_n x_0$, so $h = h_{\text{lin}} = 0$ and the same formula holds. In the reducible case, one can first restrict the system to the affine hull of its attractor and apply the theorem there.

Example 8.2 (A genuinely reducible case of the weaker condition). *In $\mathbb{R}^2$, take $g_0(x) = x/5$, $g_1(x) = x/5 + e_1$, and $g_2(x) = x/5 + e_2$, with equal probabilities. The rotations are identities, so every line is invariant. The pieces are disjoint and the attractor spans the plane, while $\dim \nu = \log 3 / \log 5 < 1$. No nonzero subspace can be saturated, since $\dim \nu_x^V \leq \dim \nu < 1 \leq \dim V$. Thus the weaker condition permits fully reducible rotations.*

Example 8.3 (Why the saturation alternative is necessary). *Let $r = 2^{-1/2}$ and consider the four planar homotheties*

$$(x, y) \mapsto r(x, y) + (a, b), \quad a, b \in \{0, 1\},$$

with independent digit weights: a is unbiased and b has probability p of being 1. Choose $p \in (0, 1)$ with $H(p) = -p \log p - (1-p) \log(1-p) < \lambda = \frac{1}{2} \log 2$. There are no exact overlaps: a polynomial with coefficients in $\{-1, 0, 1\}$ cannot vanish at r , since its primitive minimal polynomial $2t^2 - 1$ cannot divide a nonzero polynomial with leading coefficient ± 1 . Hence $h = \log 2 + H(p)$.

The stationary measure is a product. Its first factor is absolutely continuous: separating even and odd digits expresses it as the sum of two independent scaled uniform variables. The one-dimensional case of Corollary 1.3 gives dimension $H(p)/\lambda$

for the second factor. Therefore

$$\dim \nu = 1 + H(p)/\lambda < 2 = \min\{2, h/\lambda\}.$$

The horizontal direction is saturated, and its quotient entropy is $H(p)$; its term in (1.2) gives exactly the actual dimension. Thus virtual solvability and affine irreducibility alone cannot imply the unreduced formula.

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